Technical Research Report

Power Electronics and Control Systems

Single Phase Locked Loop:

structures, transfer functions and laboratory implementation.

Grid synchronization of single phase signals through SOGI-QSG and Park based PLL.

Authors:	**Revision:**	**Date:**
Davide Bagnara	00	May 2021

Abstract: The phase locked loop (*PLL*) is the standard way of extracting the phase angle, the frequency and the amplitude of a grid voltage, and it is therefore the entry point of any grid connected converter control. In a single phase system the task is harder than in a three phase one, because a single measurement does not provide the orthogonal component needed to build a rotating vector: the quadrature signal has to be generated. This document reviews the structures which solve that problem. The elementary *PLL* is introduced first (phase detector, loop filter and voltage controlled oscillator), it is linearized around the locked condition and its open and closed loop transfer functions are derived, showing the double pole at the origin which makes the loop able to track a frequency ramp with no steady state error. Two quadrature signal generators are then analysed: the second order generalized integrator (*SOGI-QSG*), whose direct, quadrature and error transfer functions behave as a band pass, a low pass and a notch filter tuned on the estimated frequency, and the scheme based on the direct and inverse Park transforms, for which the equivalent block diagram and the corresponding transfer functions are derived. Bode diagrams are reported for both. The structures are intended for a laboratory implementation on a dSPACE SCALEXIO real-time system, with the moving average filter written in C and integrated through the Simulink C-caller.

Revision history

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1 Introduction

In this document the following topics are slightly covered:

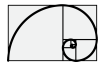
- phase locked loop for single phase signals;
- moving average filter;
- discrete Fourier transform;
- Simulink C-caller;
- dSPACE SCALEXIO.

The idea of the document is to propose an laboratory application of a single-phase *PLL*, and moving average filter implemented in a fast prototyping equipment. The moving average filter will be implemented using customized C-code and Simulink C-caller.

2 Nomenclature

Here a list of symbols, variables, and parameters used along the document:

- *PLL*: phase locked loop;
- *VCO*: voltage controlled oscillator;
- *PI*: proportional integral controller;
- *LF*: loop filter;
- *PD*: phase detector;
- *SOGI*: second order generalized integrator;
- *QSG*: quadrature signal generator;
- *MAVG*: moving average;
- RMB: right mouse button;
- LMB: left mouse button;
- HMI: human machine interface;
- ConfigurationDesk: dSPACE application used to configure a project with the scalexio equipment;
- ControlDesk: dSPACE application used as HMI;
- $v,\ v_{signal}\quad \left[\mathrm{V}\right]$: input signals;



– $\alpha\beta$: direct and quadrature components of a vector quantity, generally with respect to a stationary reference frame;

– $\xi\eta$: additional direct and quadrature components of a vector quantity, generally with respect to a rotating reference frame;

3 Basic Structure of a Phase-Locked Loop

The basic structure of the phase-locked loop (*PLL*) is shown in Figure 1. It consists of three fundamental blocks:

– *phase detector* (*PD*). This block generates an output signal proportional to the phase difference between the input signal, v_{signal}, and the signal generated by the internal oscillator of the *PLL*, v^{pll}. Depending on the type of *PD*, high frequency AC components appear together with the DC phase-angle difference signal.

– *loop filter* (*LF*). This block presents a low pass filtering characteristic to attenuate the high frequency AC components from the PD output. Typically, this block is constituted by a first order low pass filter or a PI controller.

– *voltage controlled oscillator* (*VCO*). This block generates at its output an AC signal whose frequency is shifted with respect to a given central frequency, ω_{ff}, as a function of the input voltage provided by the *LF*.

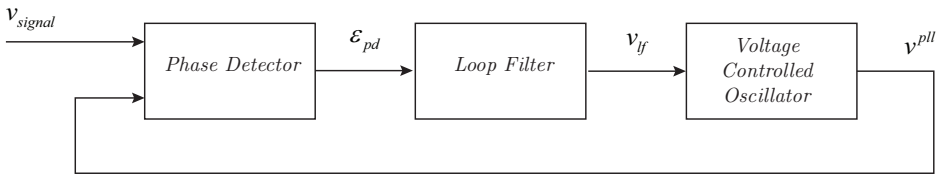


Figure 1: Basic structure of a PLL.

The block diagram of an elementary *PLL* is shown in Figure 2. In this case *PD* is implemented by means of the *Superheterodyne* technique, the *LF* is based on a *PI* controller and the *VCO* consists of a sinusoidal function supplied by a linear integrator.

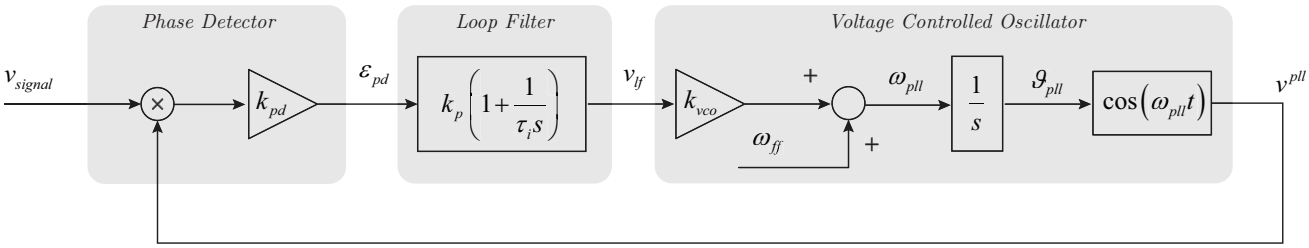


Figure 2: Block diagram of an elementary *PLL*.

If the input signal applied to this system is given by

$$v = V\sin(\vartheta) = V\sin(\omega t + \phi) \tag{3.1}$$



and the signal generated by the *VCO* is given by

$$v^{pll} = \cos(\vartheta_{pll}) = \cos(\omega_{pll}t + \phi_{pll}) \tag{3.2}$$

the phase error signal from the multiplier *PD* output can be written as

$$\begin{aligned} \varepsilon_{pd} &= Vk_{pd}\sin(\omega t + \phi)\cos(\omega_{pll}t + \phi_{pll}) \\ &= \frac{Vk_{pd}}{2}\Big[\sin[(\omega - \omega_{pll})t + (\phi - \phi_{pll})] + \sin[(\omega + \omega_{pll})t + (\phi + \phi_{pll})]\Big] \end{aligned} \tag{3.3}$$

The high frequency components $(\omega + \omega_{pll})$ of the *PD* error signal will be cancelled out by the *LF*, only the low frequency term $(\omega - \omega_{pll})$ will be processed, therefore, the *PD* error signal to be considered is

$$\varepsilon_{pd} = \frac{Vk_{pd}}{2}\sin[(\omega - \omega_{pll})t + (\phi - \phi_{pll})] \tag{3.4}$$

If it is assumed that the *VCO* is well tuned to the input frequency, i.e. with $\omega \approx \omega_{pll}$, the DC term of the phase error is given as follows

$$\varepsilon_{pd} = \frac{Vk_{pd}}{2}\sin(\phi - \phi_{pll}) \tag{3.5}$$

It can be observed in (3.5) that the multiplier *PD* produces nonlinear phase detection because of the sinusoidal function. However, when phase error is very small, i.e. when $\phi \approx \phi_{pll}$, the output of the multiplier *PD* can be linearized in the vicinity of such an operating point since $\sin(\phi - \phi_{pll}) \approx \sin(\vartheta - \vartheta_{pll}) \approx (\vartheta - \vartheta_{pll})$. Therefore, once the *PLL* is locked, the relevant term of the phase error signal is given by

$$\varepsilon_{pd} = \frac{Vk_{pd}}{2}(\vartheta - \vartheta pll) \tag{3.6}$$

According to Eq. (3.6), the model presented in Figure 2 can be linearized around the condition of $\omega \approx \omega_{pll}$ resulting as per Figure 3.

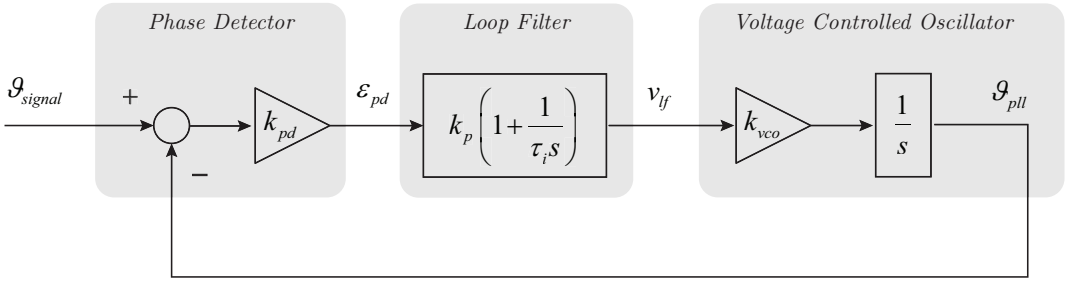


Figure 3: Small signal model of an elementary *PLL*.

According to the block diagram of Figure 3 a frequency domain analysis brings to the following transfer functions (consider $k_{pd} = k_{vco} = 1$):

$$H(s) = PD(s)LF(s)VCO(s) = \frac{k_p s + \frac{k_p}{\tau_i}}{s^2} \tag{3.7}$$

$$H_\vartheta(s) = \frac{\Theta_{pll}(s)}{\Theta(s)} = \frac{H(s)}{1+H(s)} = \frac{k_p s + \frac{k_p}{\tau_i}}{s^2 + k_p s + \frac{k_p}{\tau_i}} \tag{3.8}$$

$$E_\vartheta = \frac{E_{pd}(s)}{\Theta(s)} = 1 - H_\vartheta(s) = \frac{s^2}{s^2 + k_p s + \frac{k_p}{\tau_i}} \tag{3.9}$$

The open loop transfer function of Eq. (3.7) shows that the *PLL* has two poles at the origin, which means that is able to track even a constant slope ramp in the input phase angle without any steady state error.

4 SOGI-QSG-based PLL

Figure 4 shows the structure of a *VCO* based on *QSG*; the structure consists of an adaptive filter.

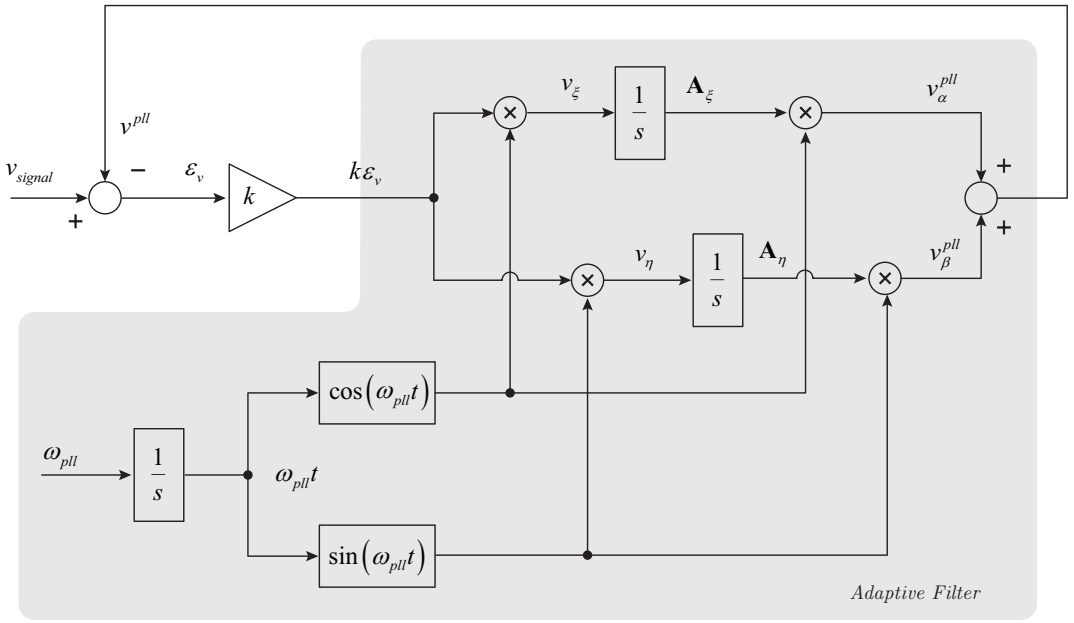


Figure 4: Voltage controlled oscillator based on a adaptive filter.

Defining $g = k\epsilon_v$, the v_ξ, and v_η components of Figure 4 can be written as follows

$$v_\xi = g\cos(\omega_{pll}) = \frac{1}{2}g\left(e^{j\omega_{pll}t} + e^{-j\omega_{pll}t}\right) \tag{4.1}$$

$$v_\eta = g\cos(\omega_{pll}) = -j\frac{1}{2}g\left(e^{j\omega_{pll}t} - e^{-j\omega_{pll}t}\right) \tag{4.2}$$

The $\mathbf{A}_\xi$, and $\mathbf{A}_\eta$ terms which correspond to the output of the integrators for v_ξ and v_η, can be expressed in the Laplace domain as follows

$$\mathbf{A}_\xi = \frac{1}{s}v_\xi(s) = \frac{1}{2s}\Big[g(s + j\omega_{pll}t) + g(s - j\omega_{pll}t)\Big] \tag{4.3}$$

$$\mathbf{A}_\eta = \frac{1}{s} v_\eta(s) = -j\frac{1}{2s}\Big[g(s+j\omega_{pll}t) - g(s-j\omega_{pll}t)\Big] \tag{4.4}$$

and the v_α^{pll}, v_β^{pll} terms result as follows

$$\begin{aligned} v_\alpha^{pll} &= \frac{1}{2}\Big[\mathbf{A}_\xi(s+j\omega_{pll}t) + \mathbf{A}_\xi(s-j\omega_{pll}t)\Big] \\ &= \frac{1}{4(s+j\omega_{pll})}\Big[g(s) + g(s+j\omega_{pll}t)\Big] + \frac{1}{4(s-j\omega_{pll})}\Big[g(s) + g(s-j\omega_{pll}t)\Big] \end{aligned} \tag{4.5}$$

$$\begin{aligned} v_\beta^{pll} &= -j\frac{1}{2}\Big[\mathbf{A}_\xi(s+j\omega_{pll}t) - \mathbf{A}_\xi(s-j\omega_{pll}t)\Big] \\ &= \frac{1}{4(s+j\omega_{pll})}\Big[g(s) + g(s+2j\omega_{pll}t)\Big] + \frac{1}{4(s-j\omega_{pll})}\Big[g(s) - g(s-2j\omega_{pll}t)\Big] \end{aligned} \tag{4.6}$$

The term $v^{pll} = v_\alpha^{pll} + v_\beta^{pll}$ results as follows

$$v^{pll} = v_\alpha^{pll} + v_\beta^{pll} = \frac{s}{s^2+\omega_{pll}^2} g(s) \tag{4.7}$$

Consequently, the transfer functions of the adaptive filter *VCO* structure of Figure 4 are given by

$$\frac{v^{pll}}{\varepsilon_v}(s) = \frac{ks}{s^2+\omega_{pll}^2} \tag{4.8}$$

$$\frac{v^{pll}}{v}(s) = \frac{ks}{s^2+ks+\omega_{pll}^2} \tag{4.9}$$

$$\frac{\varepsilon_v}{v}(s) = \frac{s^2+\omega_{pll}^2}{s^2+ks+\omega_{pll}^2} \tag{4.10}$$

The structure of Figure 4 can be used of quadrature signal generator (QSG) by adding a scaled integrator at the output of the adaptive filter, as in Figure 5 is shown.

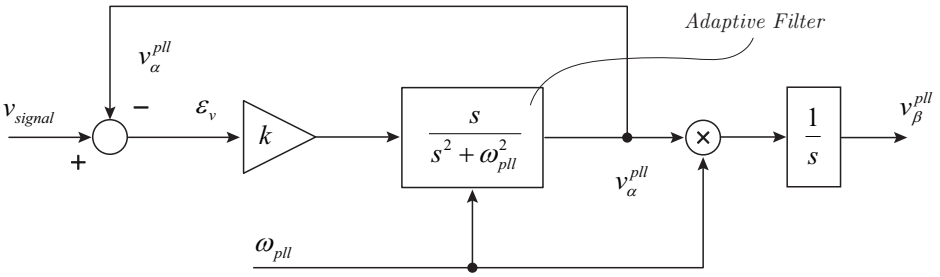


Figure 5: Quadrature signal generator based on a adaptive filter.

Clearly, the response of the AF block, shown in Figure 4 is defined by Eq. (4.8) in the case of applying a likewise sinusoidal signal (sine or cosine) with frequency ω_{pll} to its input.

Recalling that

$$\mathscr{L}\Big[\sin(\omega_{pll})\Big] = \frac{\omega_{pll}}{s^2+\omega_{pll}^2} \tag{4.11}$$

$$\mathscr{L}\Big[\cos(\omega_{pll})\Big] = \frac{s}{s^2+\omega_{pll}^2} \tag{4.12}$$

the time response of the system characterized by Eq. (4.8) in the presence of sinusoidal inputs is given by

$$\mathscr{L}^{-1}\left[\frac{\omega_{pll}}{s^2+\omega_{pll}^2}\frac{s}{s^2+\omega_{pll}^2}\right] = \frac{1}{2}t\sin(\omega_{pll}t) \tag{4.13}$$

and

$$\mathscr{L}^{-1}\left[\frac{s}{s^2+\omega_{pll}^2}\frac{s}{s^2+\omega_{pll}^2}\right] = \frac{1}{2}\left[\frac{\sin(\omega_{pll}t)}{\omega_{pll}} + t\cos(\omega_{pll}t)\right] \approx \frac{1}{2}t\cos(\omega_{pll}t) \tag{4.14}$$

An efficient implementation of the structure of Figure 5 is shown in Figure 6, where

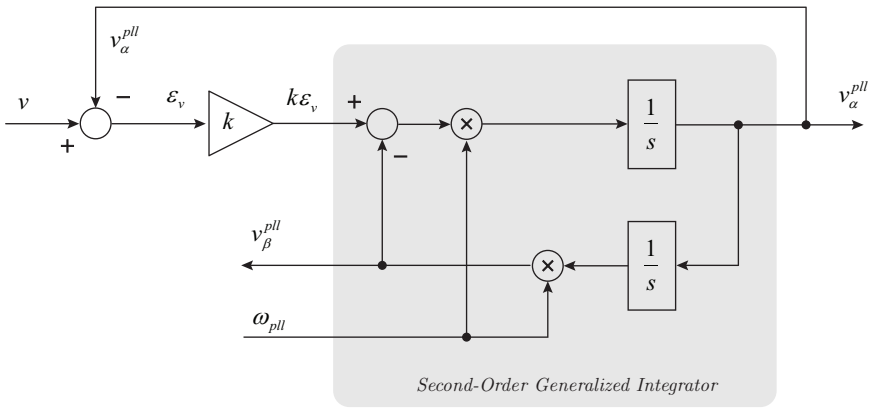


Figure 6: Second order adaptive filter based on an second order generalized integrator and a quadrature signal generator (*SOGI-QSG*).

$$SOGI(s) = \frac{v_\alpha^{pll}}{\varepsilon_v}(s) = \frac{k\,\omega_{pll}s}{s^2+\omega_{pll}^2} \tag{4.15}$$

$$D(s) = \frac{v_\alpha^{pll}}{v}(s) = \frac{k\,\omega_{pll}s}{s^2+k\,\omega_{pll}s+\omega_{pll}^2} \tag{4.16}$$

$$Q(s) = \frac{v_\beta^{pll}}{v}(s) = \frac{k\,\omega_{pll}^2}{s^2+k\,\omega_{pll}s+\omega_{pll}^2} \tag{4.17}$$

Based on the structure of the *SOGI-QSG* of Figure 6 it is possible to implement a *SOGI-QSG*-based *PLL* as shown in Figure 7.

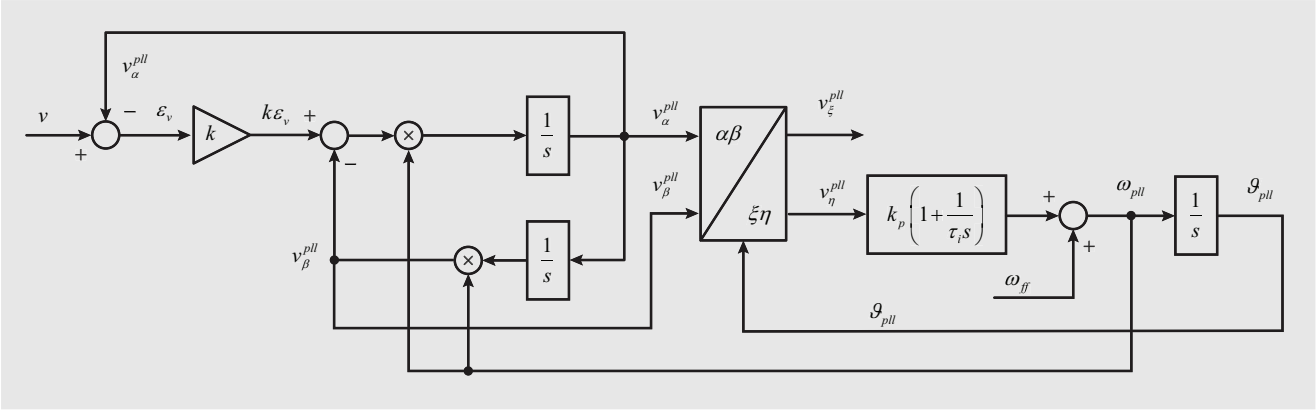
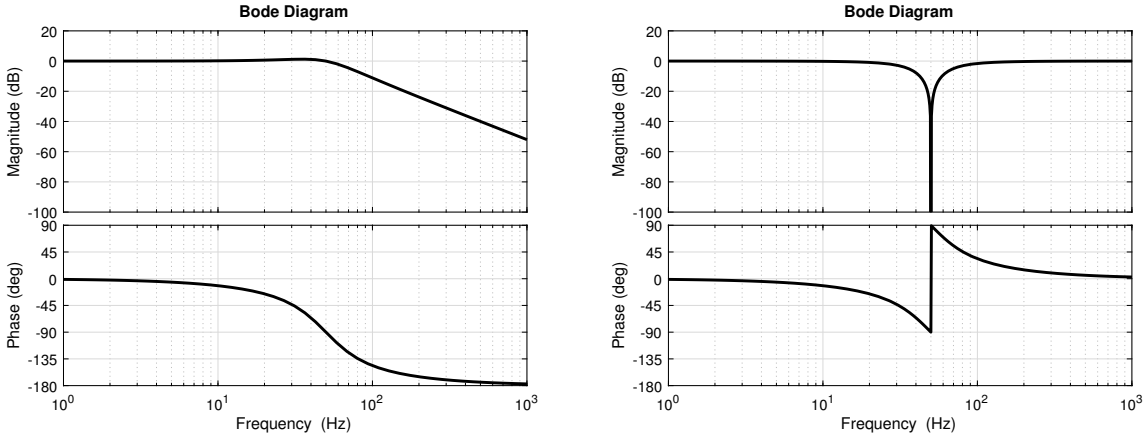


Figure 7: Diagram of the *SOGI*-based *PLL* (*SOGI-QSG*).

The transfer function from the input signal v to the error signal ε_v is given by

$$E(s) = \frac{\varepsilon_v}{v}(s) = \frac{s^2 + \omega_{pll}^2}{s^2 + k\,\omega_{pll} s + \omega_{pll}^2} \tag{4.18}$$

The transfer function of Eq. (4.18) responds to a second order notch filter, with zero gain at the centre frequency (ω_{pll}).



(a) Bode diagram of the $Q(s)$ transfer function in a *SOGI-QSG*.

(b) Bode diagram of the $E(s)$ transfer function in a *SOGI-QSG*.

Figure 8: Bode diagram of the *SOGI-QSG* response.

Another useful possible implementation of the *SOGI-QSG*-based *PLL* is based on the direct and inverse Park transforms, as shown in Figure 9

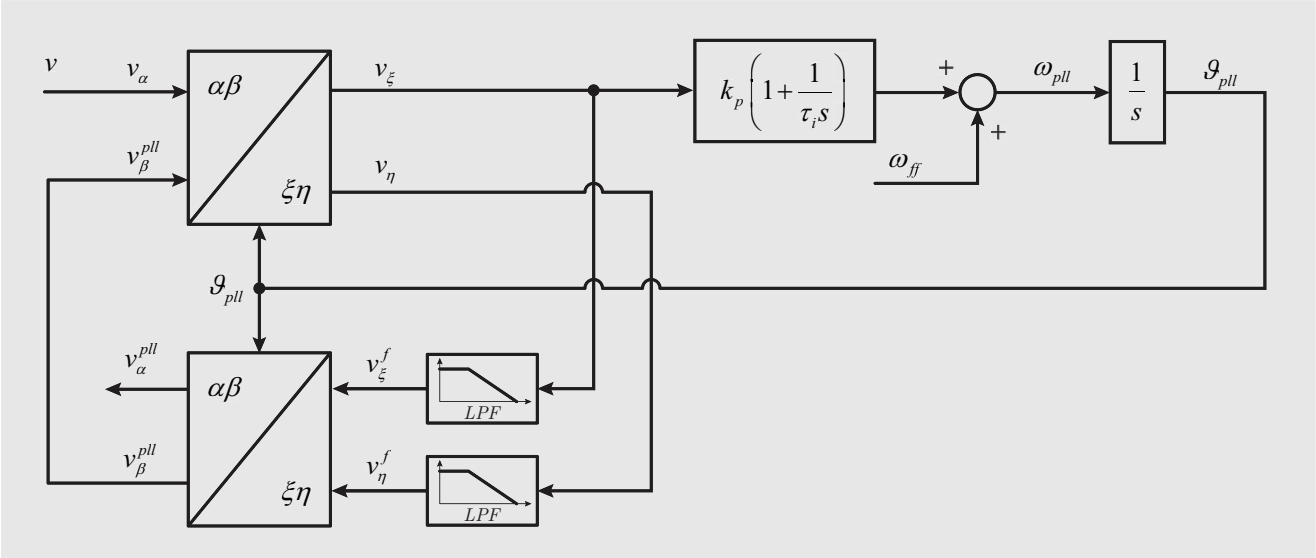


Figure 9: PLL based on the on direct and inverse Park transform.

An equivalent transfer function block diagram is presented in Figure 10. In this configuration the transformation $v_\alpha \rightarrow v_\beta^{pll}$ is represented as ω_{pll}/s in Laplace domain. An intuitive explanation of its operation principle can be given if it is assumed that the *PLL* is well tuned to the input signal frequency. Under such operation conditions, if v_α and v_β^{pll} are not in quadrature, the virtual input vector, v_α^{pll}, resulting from these signals will have neither constant amplitude nor rotation speed.

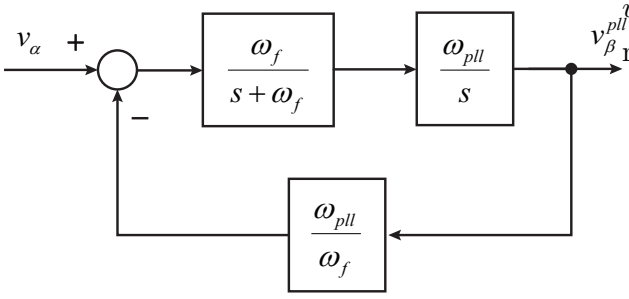


Figure 10: Equivalent block diagram of the PLL based on the on direct and inverse Park transform.

Therefore v_ξ and v_η waveform resulting from direct Park transformation will have harmonics components. These harmonics will be suppressed by the *LPF* blocks generating, as well, the components v_ξ^f and v_η^f. The v_α and v_β^{pll} components resulting from the inverse Park transformation of the components v_ξ^f and v_η^f will be in quadrature, though v_α and v_α^{pll} will not be in phase if the *PLL* is not perfectly synchronized. As the *PLL* locks the phase angle of the input signal v_α, the components v_α^{pll} and v_β^{pll} will be respectively in phase and in quadrature respect to the input signal v_α.

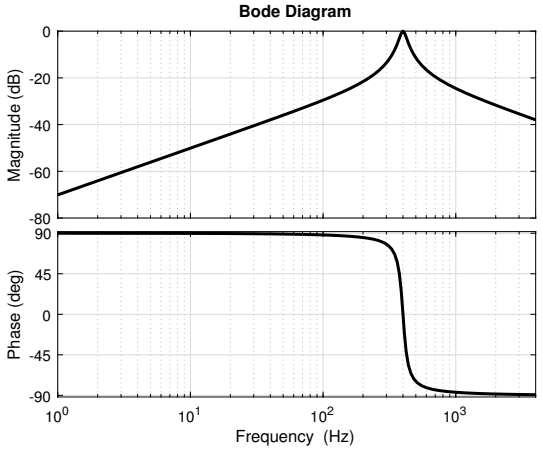
From the equivalent block diagram of Figure 10 the following transfer functions can be derived

$$\boxed{\frac{V_\beta^{pll}}{V_\alpha}(s) = \frac{\omega_f \omega_{pll}}{s^2 + \omega_f s + \omega_{pll}^2}} \tag{4.19}$$

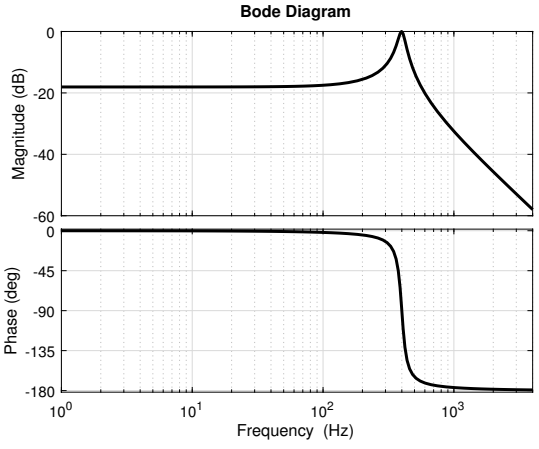
$$\boxed{\frac{V_\alpha^{pll}}{V_\alpha}(s) = \frac{s\omega_f}{s^2 + \omega_f s + \omega_{pll}^2}} \tag{4.20}$$

where the relation of Eq. (4.21) has been applied.

$$V_\beta^{pll}(s) = \frac{\omega_{pll}}{s} V_\alpha^{pll}(s) \tag{4.21}$$



(a) Bode diagram of the $\frac{V_\alpha^{pll}}{V_\alpha}(s)$ transfer function, with $\omega_f = 2\pi 50\,\text{Hz}$, and $\omega_{pll} = 2\pi 400\,\text{Hz}$.



(b) Bode diagram of the $\frac{V_\beta^{pll}}{V_\alpha}(s)$ transfer function, with $\omega_f = 2\pi 50\,\text{Hz}$, and $\omega_{pll} = 2\pi 400\,\text{Hz}$.

Figure 11: Bode diagram of the *QSG-PLL* based on direct and inverse Parck transform.

References

[1] R. Teodorescu, M. Liserre, P. Rodrıguez, *Grid converters for photovoltaic and wind power systems*. Wiley, 2011.